

Languages & Cultures Across the Curriculum

991: World Languages

Languages & Cultures Across the Curriculum Data Science in Spanish taught in Spanish

Attached to 01:960:142 - Data 101: Basic data science

Module number: 01:991:320 (1 credit)

Spring 2020

Instructor / Professor : Joseph V. Casillas

Class Hours & Location:

Office Hours:

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Languages & Cultures Across the Curriculum DESCRIPTION

This module is attached to Data 101: Basic data science (01:960:142). In this course students examine the fundamental principles of Data Science with special focus on developing practical knowledge of and experience with current tools used to explore, tidy, visualize, and analyze data. The course also teaches students how to make the aforementioned procedures reproducible and shareable. Students will develop a foundation in programming in R, as well as learning the most common tools at the disposal of today's data scientist (i.e. GitHub, Knitr, etc.). This course is designed for advanced L2 learners and heritage speakers.

I. MODULE OBJECTIVES:

Data Science in Spanish aims to help students learn/think about and appreciate data in new ways, using the Spanish language. At the end of the semester students will be able create data visualizations and complete basic analyses in a way that is reproducible and shareable, as well as connect the processes and outcomes of learning a programming language in relation to second language learning in their academic and personal lives (LE2.2). To this end students will also understand some aspects of the processes of second language learning and be able to compare and contrast them with learning a programming language (LE2.3).

Over the course of the semester, your objectives will be to:

- Understand open and reproducible science
- Learn how to get and tidy data
- Create elegant data visualizations
- Produce shareable data reports in a variety of formats
- Implement a plan for version control

II. REQUIREMENTS

Students will need access to a personal laptop computer. All materials and software are open source and freely available. This course will meet once per week for 14 weeks. Students should prepare to spend 2 hrs. per week of preparation time on the course material outside of class.

III. ASSESMENT

This module intends to help empower students with the ability to learn another language from the elementary level to the most advanced. The purpose is to develop students' skills in communication across ethnic, ideological and national boundaries, as well as help students acquire a deeper understanding of other

cultures. This undergraduate module will offer students studying statistics the opportunity to develop specialized proficiency in speaking, comprehension, reading and writing about Data Science in Spanish. This module will assess students based on their ability to demonstrate proficiency in interpretive, interpersonal and presentational modes of communication and basic knowledge of Data Science.

Weekly assignments will consist of readings and code walkthroughs. Class time will be dedicated to lecture, workshops and friendly contests. In each of the data contests, students will work in small groups and attempt to solve a data science problem using the skill they learned in the previous workshops. For the final group presentation students will use a publicly available dataset and attempt to improve upon the available analyses, plots, etc., and present their findings and a reflection of their experience to the class. More details are provided on Canvas.

CALENDAR (DUE DATES/ ASSESSMENT AND/OR CONTENT):

Calendar

Week	Topic	Task and Assessment Due
1	INTRODUCTION TO THE COURSE What is data? What is data science?	Language Engagement- Entry Survey Selected readings
2	Open science and reproducible research	Selected readings
3	Data visualization I: what makes a good dataviz?	Selected readings, activities and workshop
4	Data visualization II: intro to ggplot2	Selected readings, workshop and assignment
5	Data visualization III: ggplot2 cont.	Data visualization contest
6	Reproducible reports I: Rmarkdown	Selected readings, workshop
7	Reproducible reports II: Tips and trick	Workshop, report contest
8	Getting and cleaning data I: how not to get your hands dirty	Selected readings, workshop with dplyr
9	Getting and cleaning data II: advanced wrangling	Cleaning contest
10	Working with data I: tidying up	Selected readings, workshop
11	Working with data II: how tidy is too tidy?	Bring your dirty data day
12	Keeping it safe I: version control in the 21st	Selected readings
13	Keeping it safe II: git out here! Intro to github	Selected readings, workshop
14	Putting it all together: becoming a data scientist	END OF COURSE- Presentations Language Engagement- Exit Survey
Note: Specific details regarding assignments and contests will be available on Canvas.		

IV. GRADE DISTRIBUTION. The grade will be based on the following evaluations:

TASKS	
Participation	30%
Workshop products and assignments - Dataviz assignment a la Simon says - Generating reproducible reports in multiple formats	40%
Programming team contests - Dataviz skills challenge - Reproducible reports in the wild - Can you clean my data? - Can you clean their data?	20%
Final presentation	10%

V. STUDENTS WELLNESS SERVICES & POLICIES

ACCOMMODATIONS and ASSISTANCE: Any student who feels that s/he may need an accommodation based on a disability should bring the proper documentation from the ODS Office (<http://ods.rutgers.edu>) and discuss his or her specific needs privately with the instructor. Any student dealing with extraordinary personal situation should contact (<http://rhscaps.rutgers.edu/>).

ATTENDANCE: If you are going to miss a class or evaluation, within 24 hours of your absence you are **required** to use the University absence reporting website <https://sims.rutgers.edu/ssra/> to indicate the date and reason for your absence. Please keep in mind that using the reporting system does not mean that your absence is automatically considered “excused,” nor that you will be offered a make-up. In some instances you may need to get further documentation excusing your absence. If you must miss a class or exam for religious reasons, please consult <http://www.state.nj.us/education/genfo/holidays1718.pdf>. It is your responsibility to inform your absence..

ACADEMIC HONESTY: Students are not allowed to use computer-assisted translation software or seek outside help in any form. In other words, the writing is to be entirely your own work (words and ideas). If you are using material produced by another person under any circumstances, you must acknowledge it. Any outside help is considered unacceptable collaboration and will be dealt with accordingly. You are responsible for reviewing the University’s definitions and policies for plagiarism and academic dishonesty (<http://academicintegrity.rutgers.edu/academic-integrity-at-rutgers>).

- **Plagiarism** is the representation of the words or ideas of another as one's own in any academic exercise. To avoid plagiarism, every direct quotation must be identified by quotation marks or by appropriate indentation and must be properly cited in the text or in a footnote. Acknowledgment is required when material from another source stored in print, electronic or other medium is paraphrased or summarized in whole or in part in one's own words. To acknowledge a paraphrase properly, one might state: 'to paraphrase Plato's comment...' and conclude with a footnote identifying the exact reference. A footnote acknowledging only a directly quoted statement does not suffice to notify the reader of any preceding or succeeding paraphrased material. Information which is common knowledge such as names of leaders of prominent nations, basic scientific laws, etc., need not be footnoted; however, all facts or information obtained in reading or research that are not common knowledge among students in the course must be acknowledged.

A student who hands in work that is totally or partially plagiarized will be reported to the Office of Student Judicial affairs. Any questions about what constitutes plagiarism should be discussed with the faculty member.